

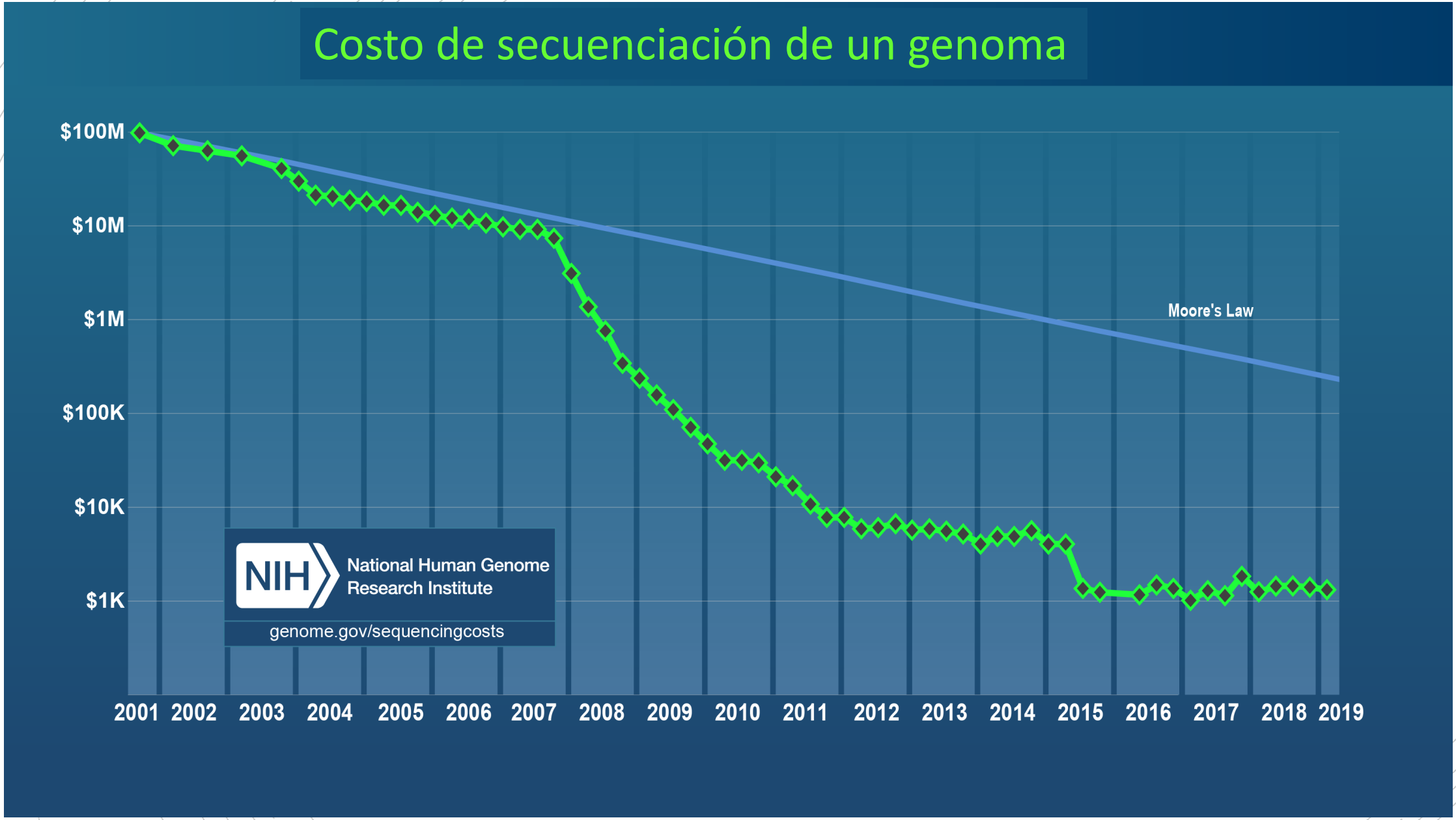
Introducción a la Biotecnología 2026

CRISPR & RNAi

Temario

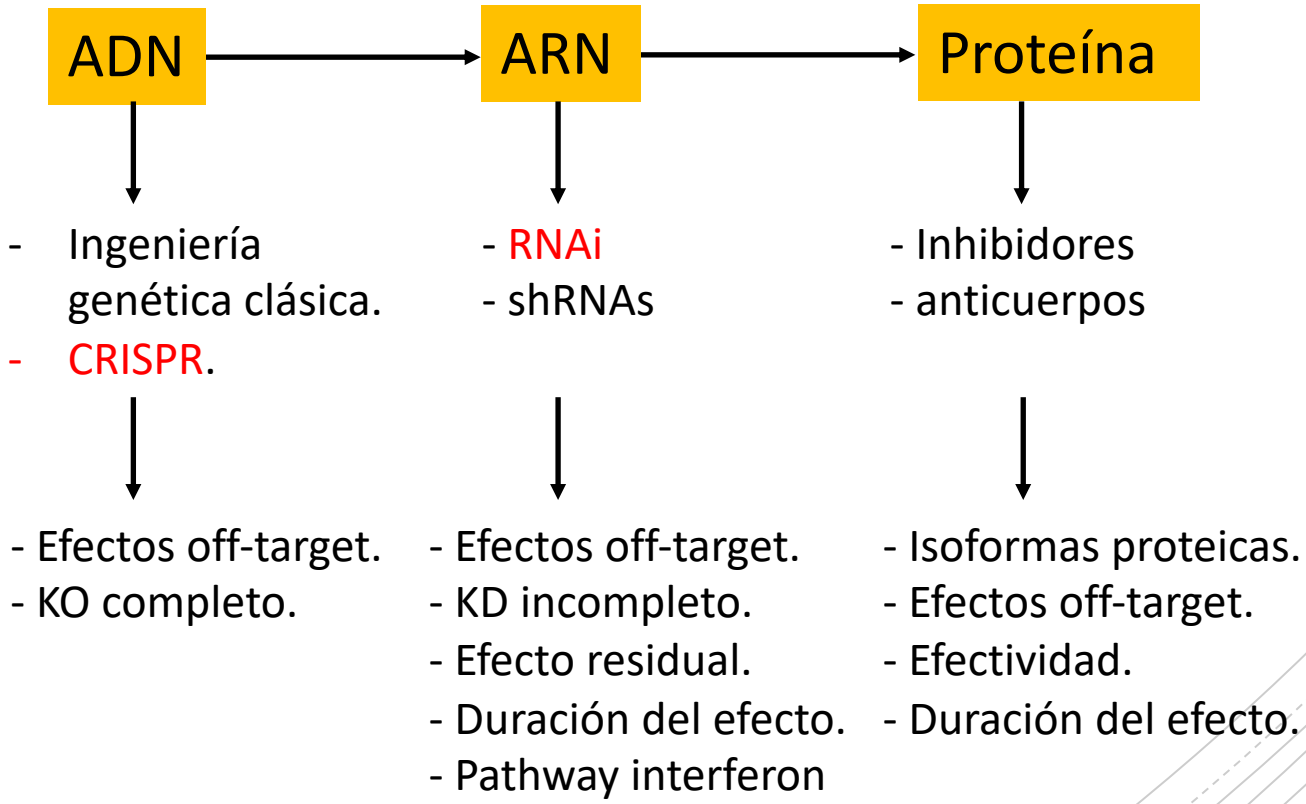
- Contexto histórico / Experimentos de perdida de función.
- ARNi, descubrimiento, maquinaria molecular involucrada.
- CRISPR, descubrimiento como mecanismo de inmunidad en bacterias.
- CRISPR, uso como herramienta de edición génica.
- Detección de la edición en sistemas heterólogos.
- Otros usos de CRISPR.
- Diferencias CRISPR versus RNAi.

Costo de secuenciación de un genoma

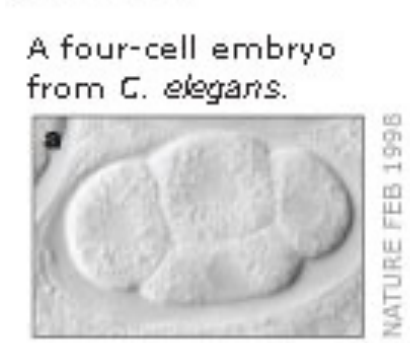


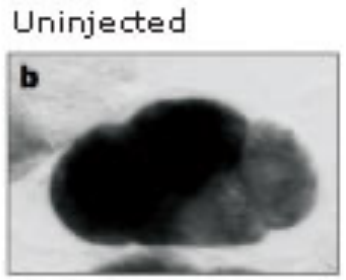
¿Cuál es la función del gen X?

- Experimentos funcionales: localización, interacciones.
- Gold Standard: Remover función del gen X: ¿que pasa en la célula/modelo animal sin este gen?

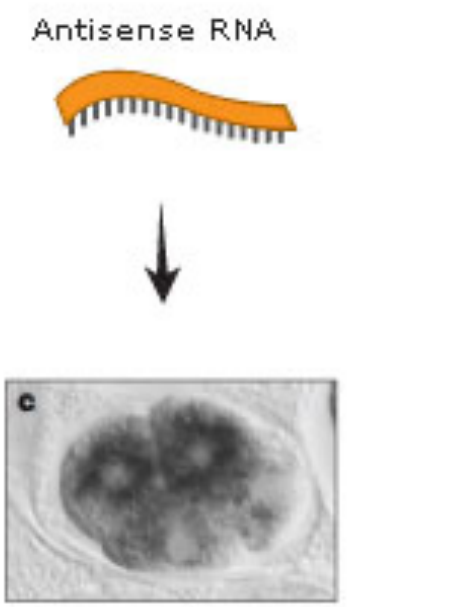


ARN de interferencia (RNAi)

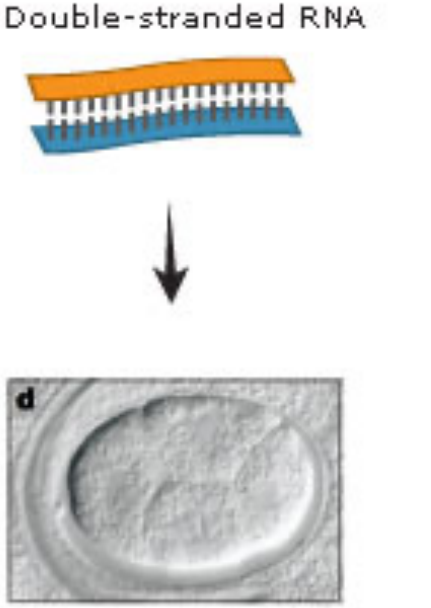




mex-3 RNA (stained black) is abundant in the early embryo.



Injection of antisense RNA reduced the content of mRNA to some extent.



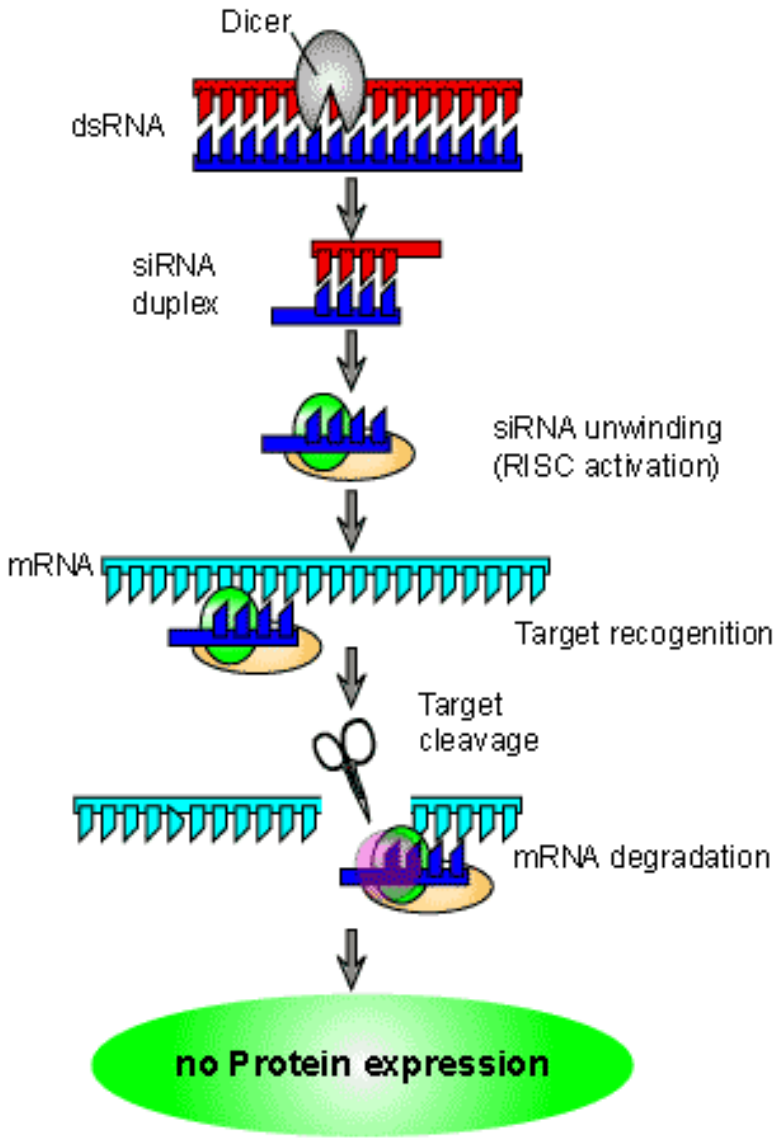
The target mRNA was eliminated after injection of double-stranded RNA.

https://www.youtube.com/watch?v=wRzHj2w8S2I
https://www.youtube.com/watch?v=cK-OGB1_ELE

ARN de interferencia (RNAi)

- Mecanismo regulatorio de la expresión de genes que limita los niveles de transcripto.
- a) Supresión de la transcripción (silenciamiento transcripcional de genes).
- b) Activación de la degradación del RNA de manera secuencia-específica (silenciamiento post-transcripcional de genes/interferencia de RNA).

ARN de interferencia (RNAi)



PASO 1 : Iniciación

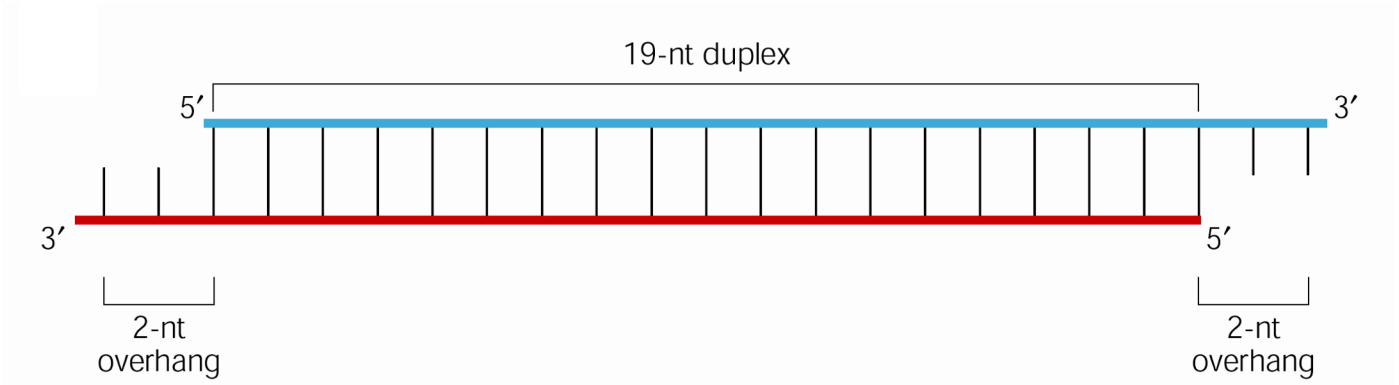
- Unión de Dicer a un dsRNA (30-200 nt).
- Clivaje en fragmentos de RNA (siRNA) de 21-25 nt.

PASO 2 : Efector

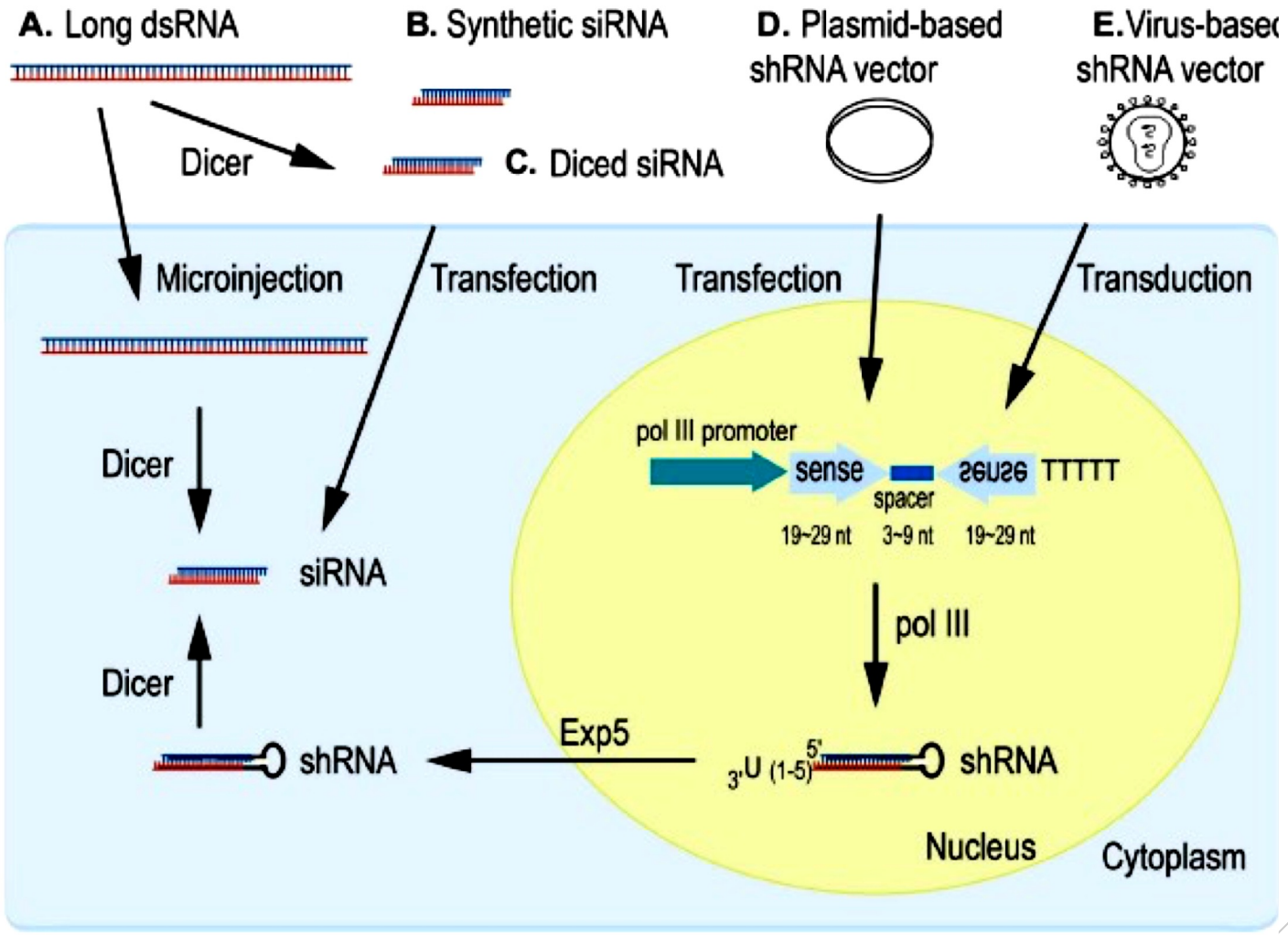
- siRNAs se unen al complejo RISC.
- Degradación de mRNA simple cadena homólogo.

Estructura de un siRNA

- extremo 5’-fosfato
- extremo 3’-hidroxilo
- 2/3-nucleótidos protruyentes en el extremo 3’



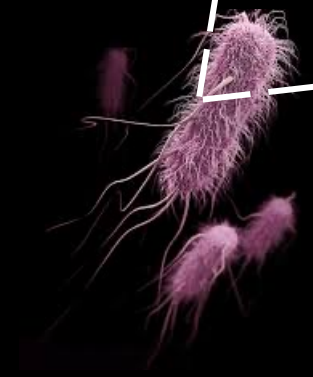
ARN de interferencia (RNAi)



CRISPR

E. Coli (1987)
Ishino et al.

GTTATGTATGCTTAGTGTGTTCCCCGCGCCAGCGGGGATAAACCGATGATCCAGCGCGGCGGCTC
ATATGAAACTGGGTGTTCCCCGCGCCAGCGGGGATAAACCGTGCGTGTCGGGCTGATTCGTCAGC
TTCTTGGTGTGTTCCCCGCGCCAGCGGGGATAAACCGAACGACGCACAGGATATAATGCTTGGCC
TGGGGTGTTCCCCGCGCCAGCGGGGATAAACCGTGACCGCCGATACGTTTGCTGGCGCGGCAGAA
GTGTTCCCCGCGCCAGCGGGGATAAACCGGTTACCGCGACTCATGCGACGATAAAAAATACGTGT
TCCCCGCGCCAGCGGGGATAAACCGGTGCCGCGACTCACCAGATAGAAATAACGCAAGTGTTCCC
CGCGCCAGCGGGGATAAACCGCCACATCCAGCATTTGCTGAGGTGAAATCCAGGTGTTCCCCGCG
CCAGCGGGGATAAACCGTTAACTTCCTGCGTCTGCTTGGGGGAATGGCCGTGTTCCCCGCGCCAG
CGGGGATAAACCGTTAGCGTGTGATTTTCCGTGTTATAGGTTAGCGTGTTCCCCGCGCCAGCGGG
GATAAACCGGCGGGATTGTTCCGTTTGCCCGCGCCACCAGCGTGTTCCCCGCGCCAGCGGGGATA
AACCGGGGCCGACGGTCGCAGTGCTGGACCATTTCAAGTGTTCCCCGCGCCAGCGGGGATAAACC
G

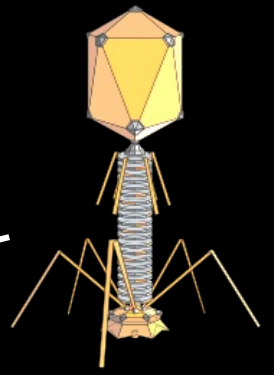


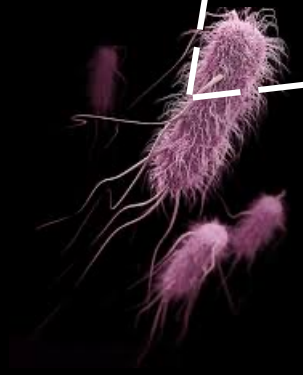
Clustered Regularly Interspaced Short Palindromic Repeats (CRISPR)
Algo así como……
Repeticiones Palindrómicas Cortas Agrupadas Intercaladas Regularmente

CRISPR

E. Coli (1987)
Ishino et al.

GTTATGTATGCTTAGTGTGTTCCCCGCGCCAGCGGGGATAAACCGATGATCCAGCGCGGCGGCTC
ATATGAAACTGGGTGTTCCCCGCGCCAGCGGGGATAAACCGTGCGTGTCGGGCTGATTCGTCAGC
TTCTTGGTGTGTTCCCCGCGCCAGCGGGGATAAACCGAACGACGCACAGGATATAATGCTTGGCC
TGGGGTGTTCCCCGCGCCAGCGGGGATAAACCGTGACCGCCGATACGTTTGCTGGCGCGGCAGAA
GTGTTCCCCGCGCCAGCGGGGATAAACCGGTTACCGCGACTCATGCGACGATAAAAAATACGTGT
TCCCCGCGCCAGCGGGGATAAACCGGTGCCGCGACTCACCAGATAGAAATAACGCAAGTGTTCCC
CGCGCCAGCGGGGATAAACCGCCACATCCAGCATTTGCTGAGGTGAAATCCAGGTGTTCCCCGCG
CCAGCGGGGATAAACCGTTAACTTCCTGCGTCTGCTTGGGGGAATGGCCGTGTTCCCCGCGCCAG
CGGGGATAAACCGTTAGCGTGTGATTTTCCGTGTTATAGGTTAGCGTGTTCCCCGCGCCAGCGGG
GATAAACCGGCGGGATTGTTCCGTTTGCCCGCGCCACCAGCGTGTTCCCCGCGCCAGCGGGGATA
AACCGGGGCCGACGGTCGCAGTGCTGGACCATTTCAAGTGTTCCCCGCGCCAGCGGGGATAAACC
G

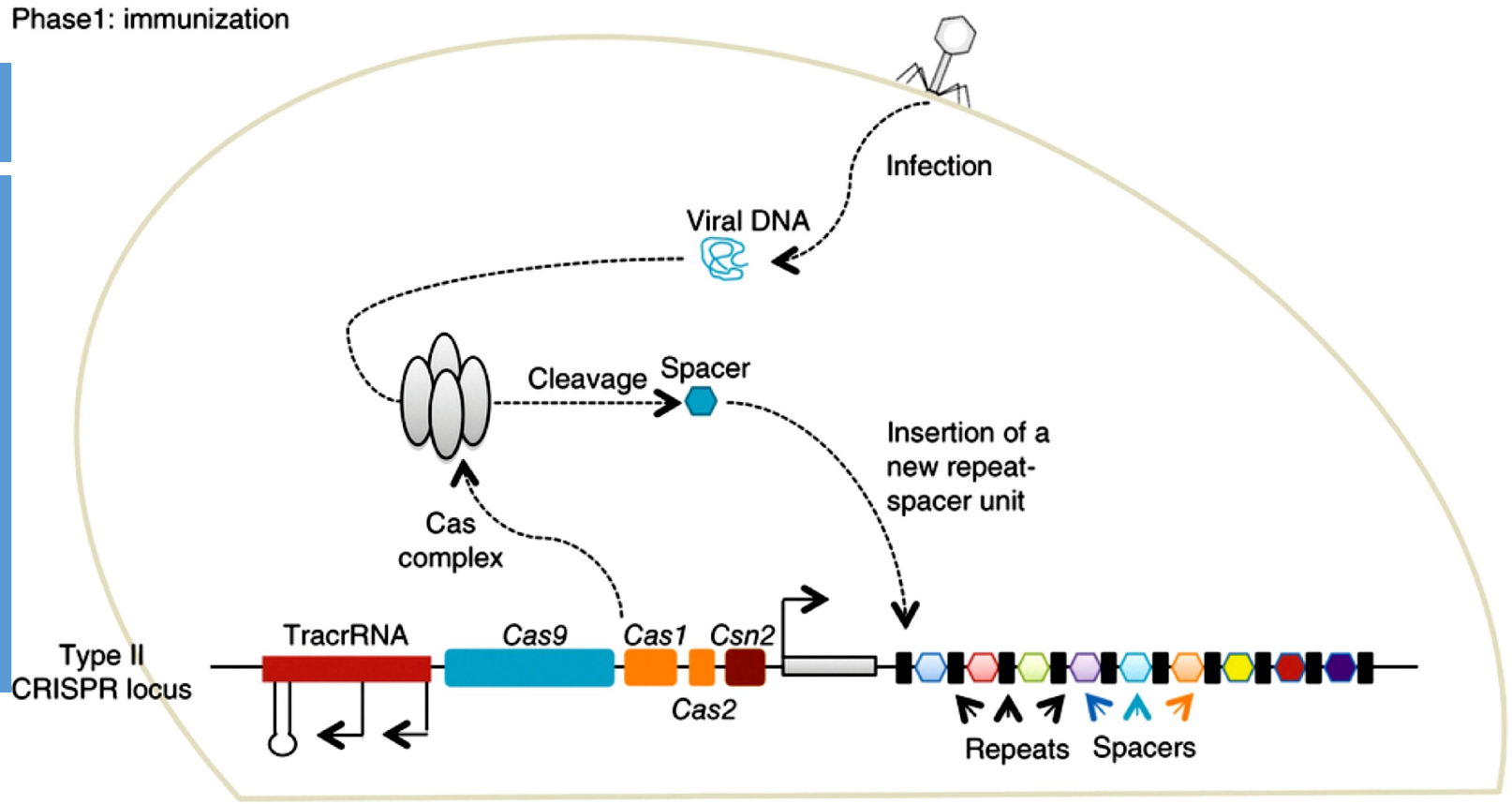




Clustered Regularly Interspaced Short Palindromic Repeats (CRISPR)
Algo así como……
Repeticiones Palindrómicas Cortas Agrupadas Intercaladas Regularmente

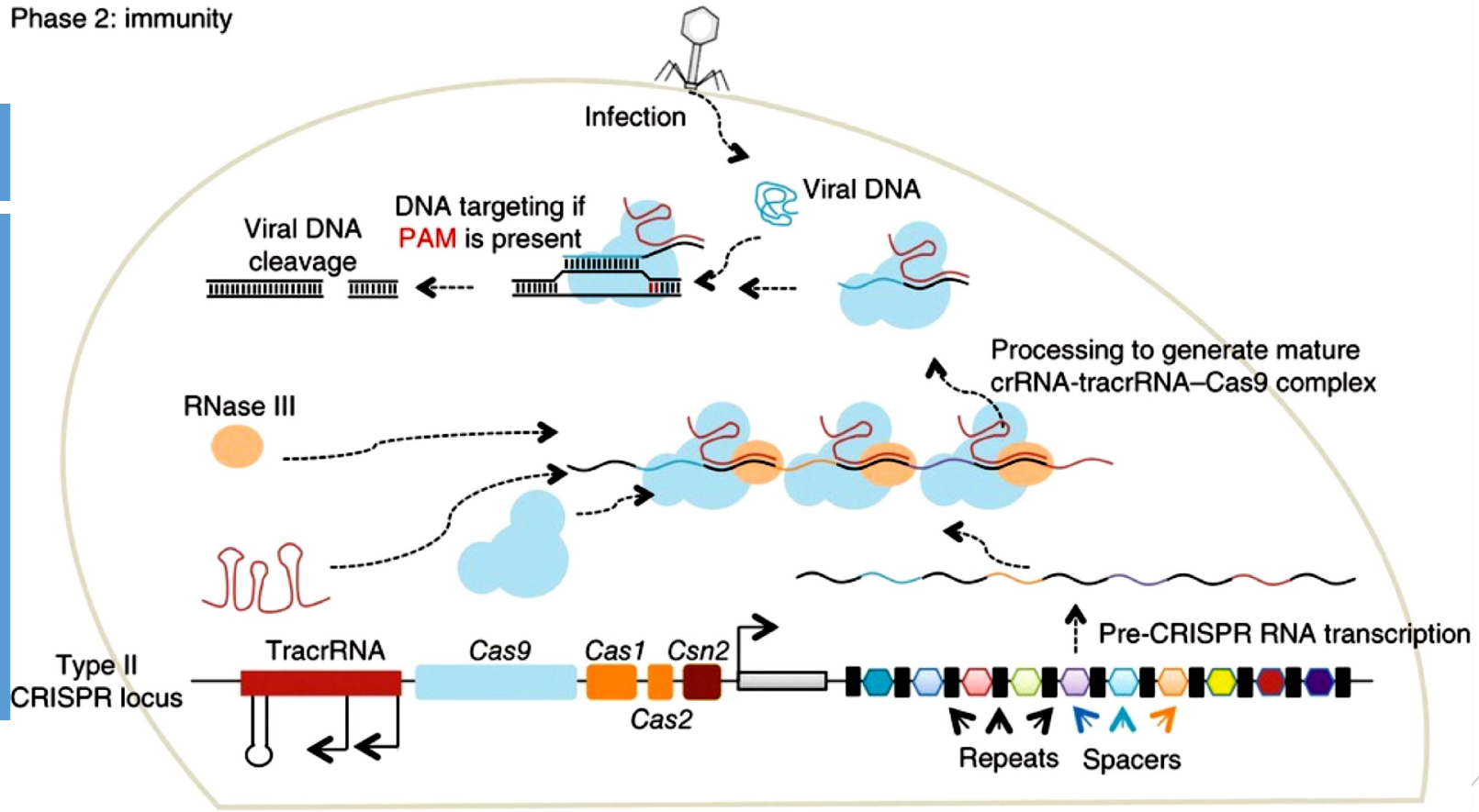
Clustered Regularly Interspaced Short Palindromic Repeats

¿Qué es CRISPR?

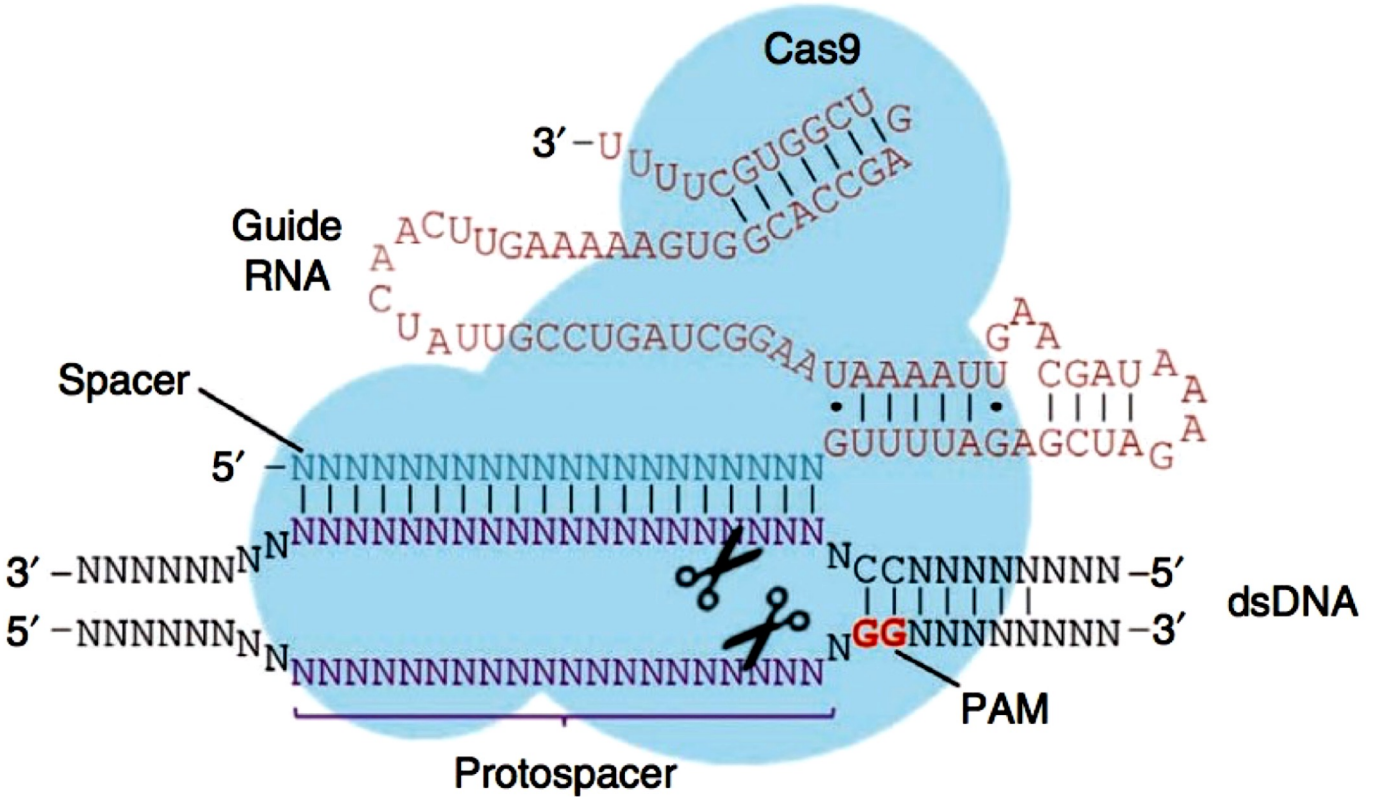


Clustered Regularly Interspaced Short Palindromic Repeats

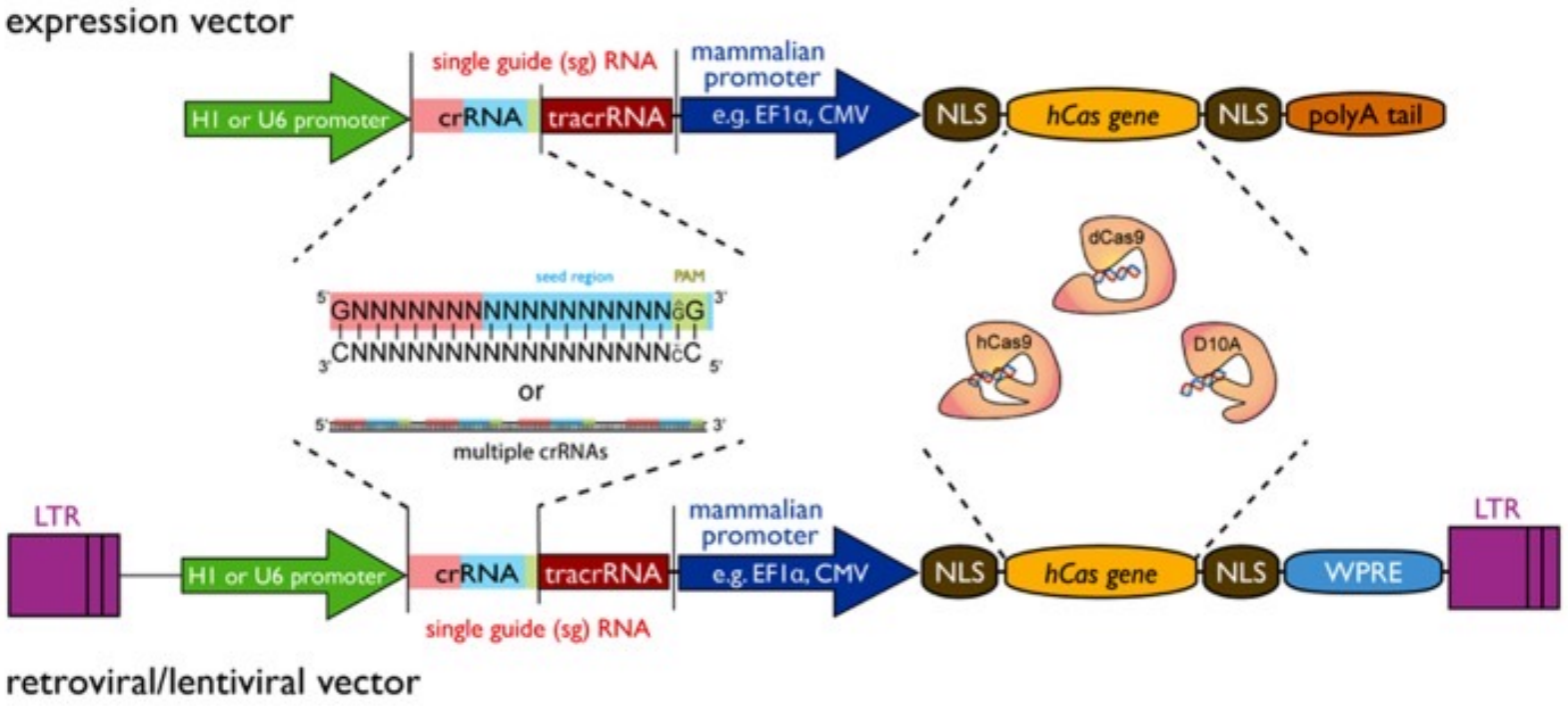
¿Qué es CRISPR?



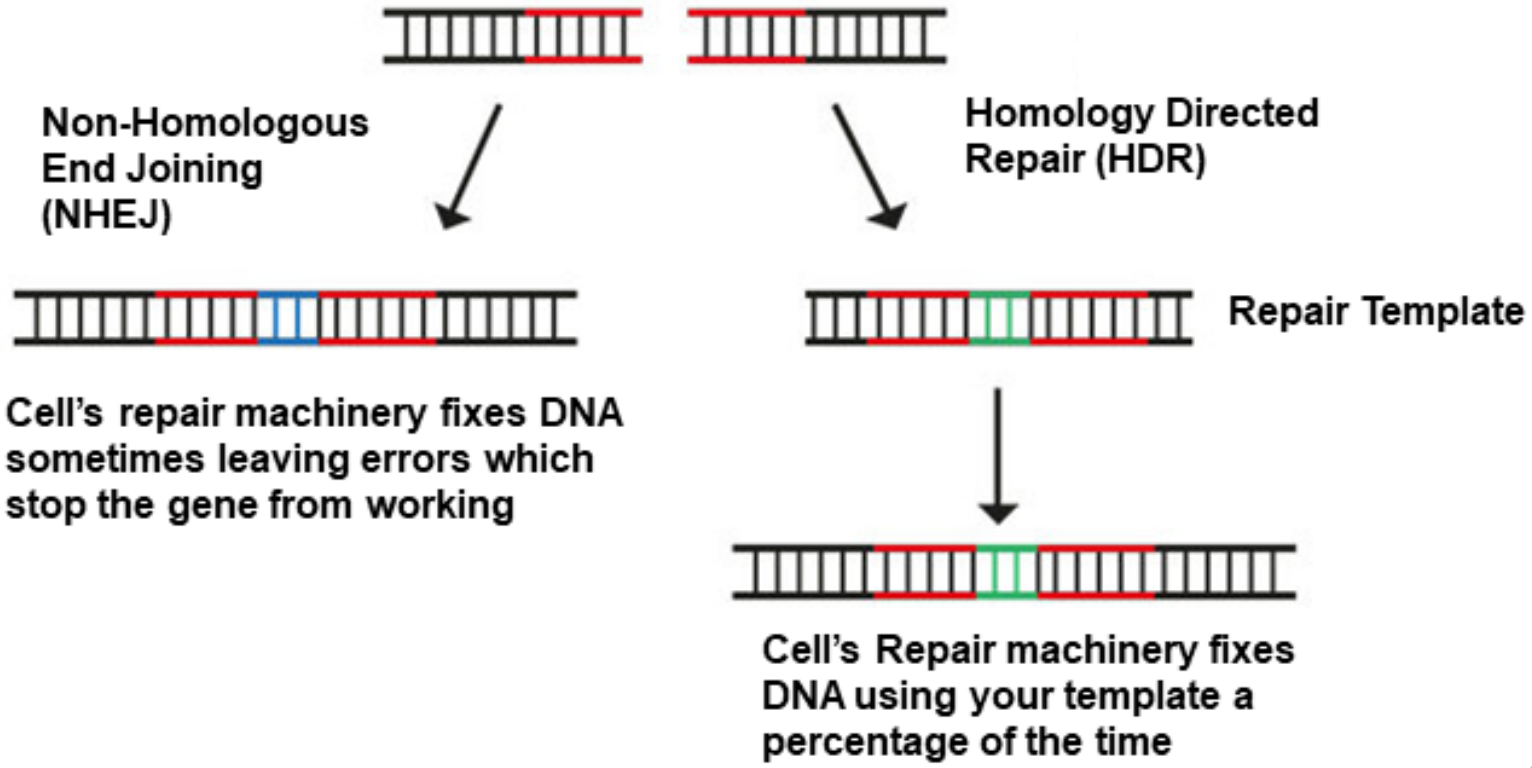
CRISPR targeting



Sistemas de edición CRISPR



Reparación del corte de Cas9



Estrategias de edición con Cas9

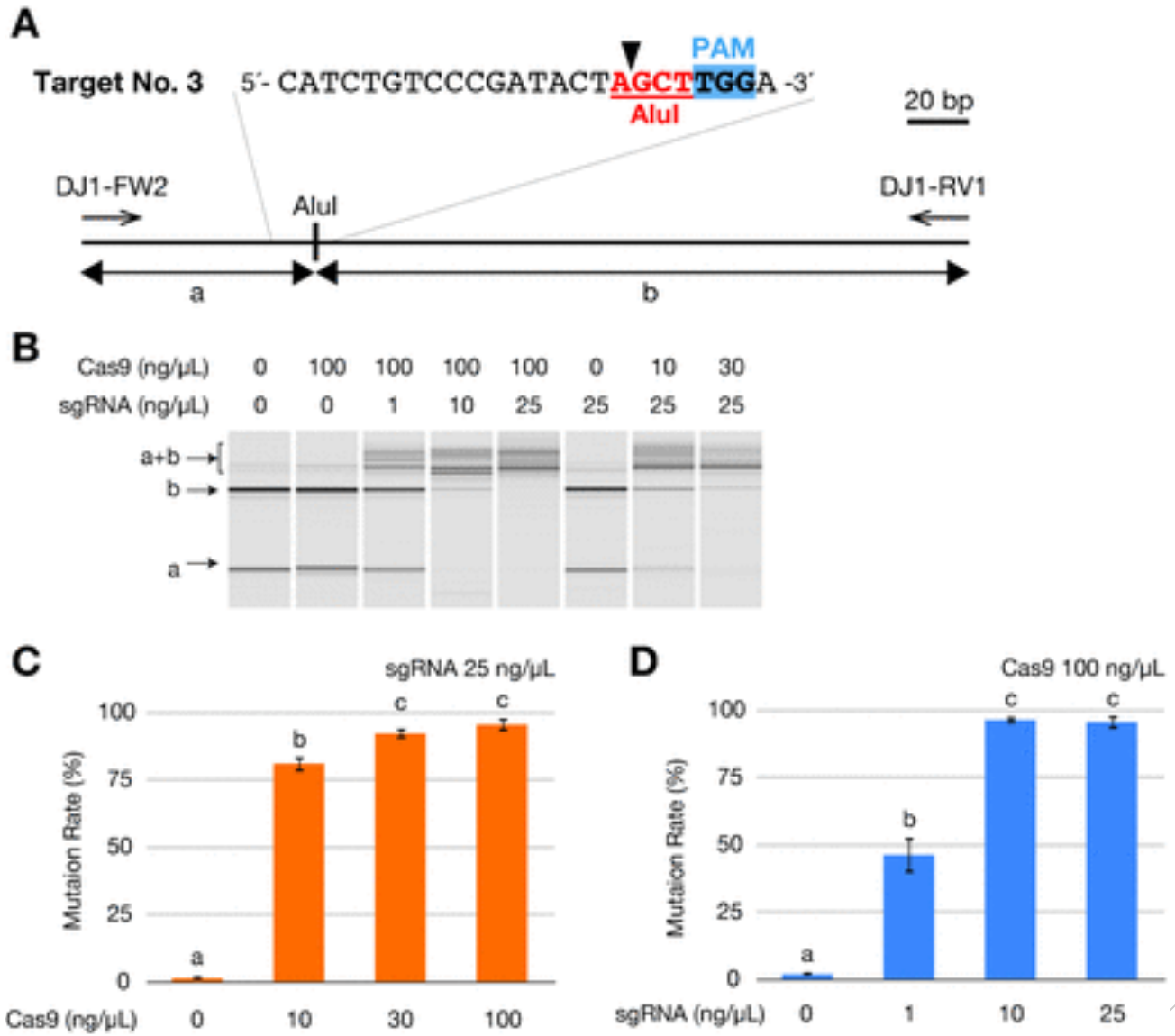
#cortes	#guías	Donante?	vía	propósito	resultado
1	1	no	NHEJ	KO	mezcla de indels
0	1	no	NMR/BER	edición de una base	mutación de transición
2	2	no	NHEJ	deleción	deleción
1	1	si	HR	KI	inserción/mutación puntual
1	1	si	NHEJ	KI	inserción grande (HITI)

Diseño experimentos CRISPR

- https://zlab.bio/guide-design-resources
- http://crispor.tefor.net/
- http://www.broadinstitute.org/rnai/public/analysis-tools/sgrna-design
- https://www.addgene.org/crispr/
- https://www.benchling.com/

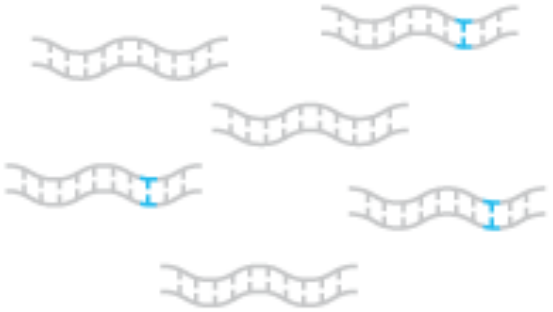
diagnóstico de edición: RFLP

Restriction fragment length polymorphism



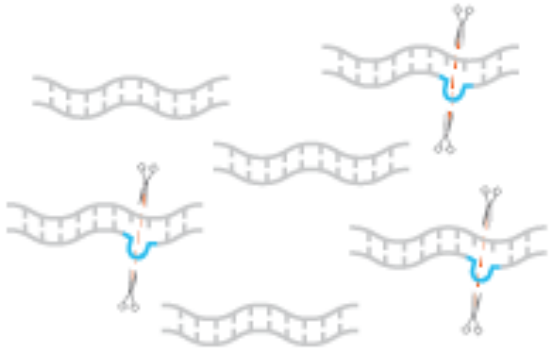
diagnóstico de edición: SURVEYOR

1 **Amplify genomic DNA from cells using Terra PCR Direct Polymerase**



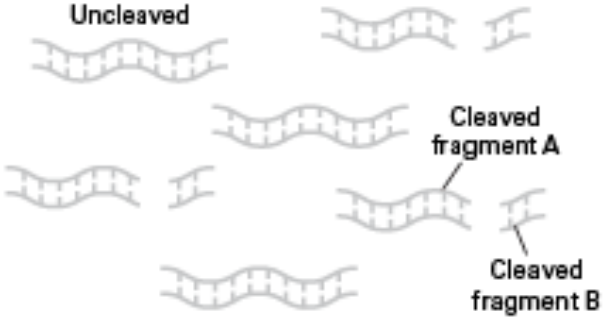
Region of interest is amplified (mutation is marked in blue)

2 **Denature and reanneal**

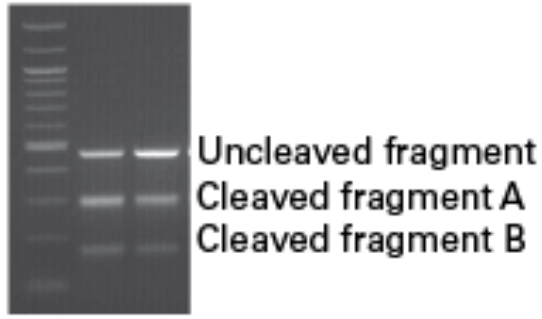


Guide-it Resolvase cleaves imperfectly matched DNA

3 **Imperfectly matched DNA is cleaved by Guide-it Resolvase**

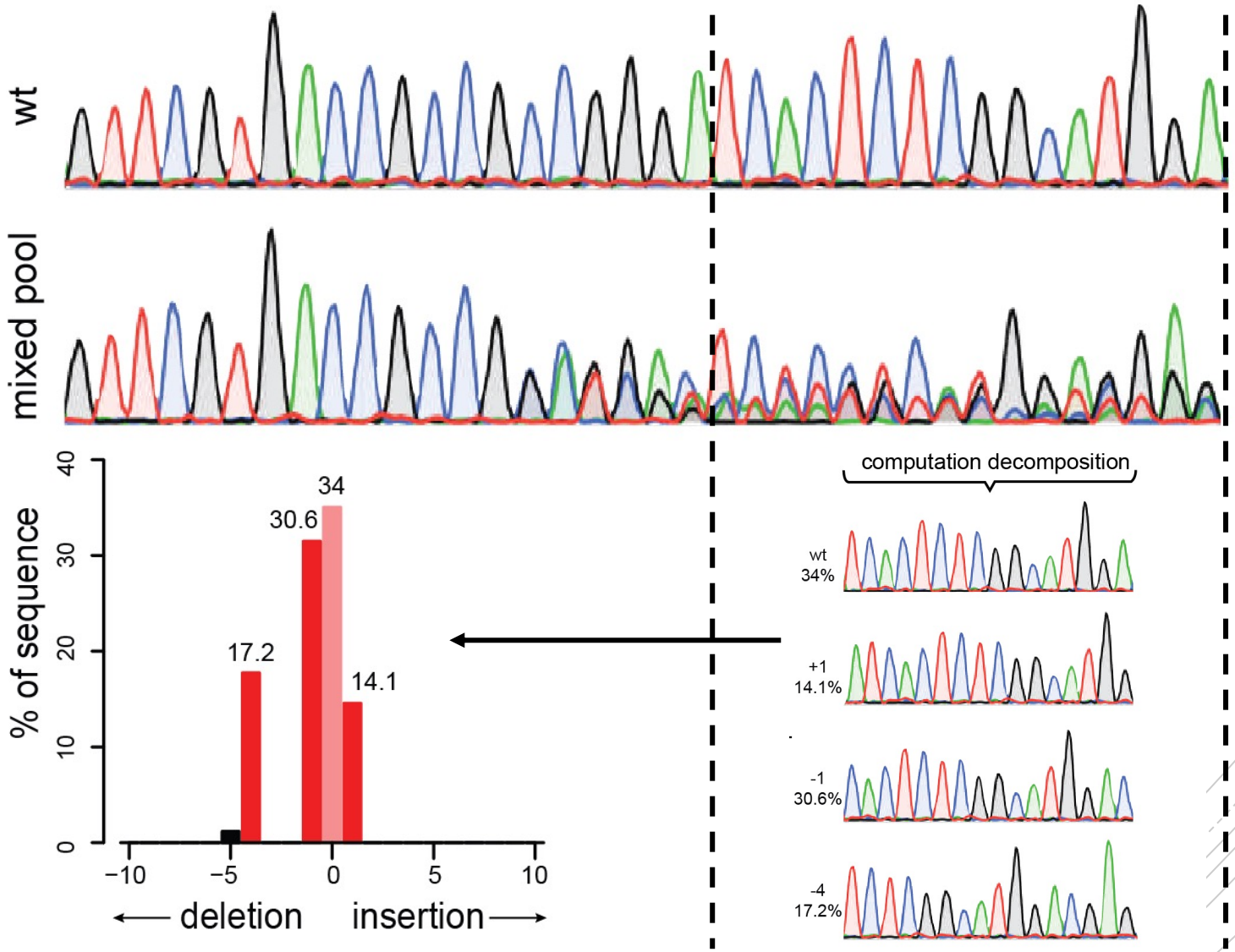


4 **Separate cleaved and uncleaved PCR fragments using agarose gel electrophoresis**

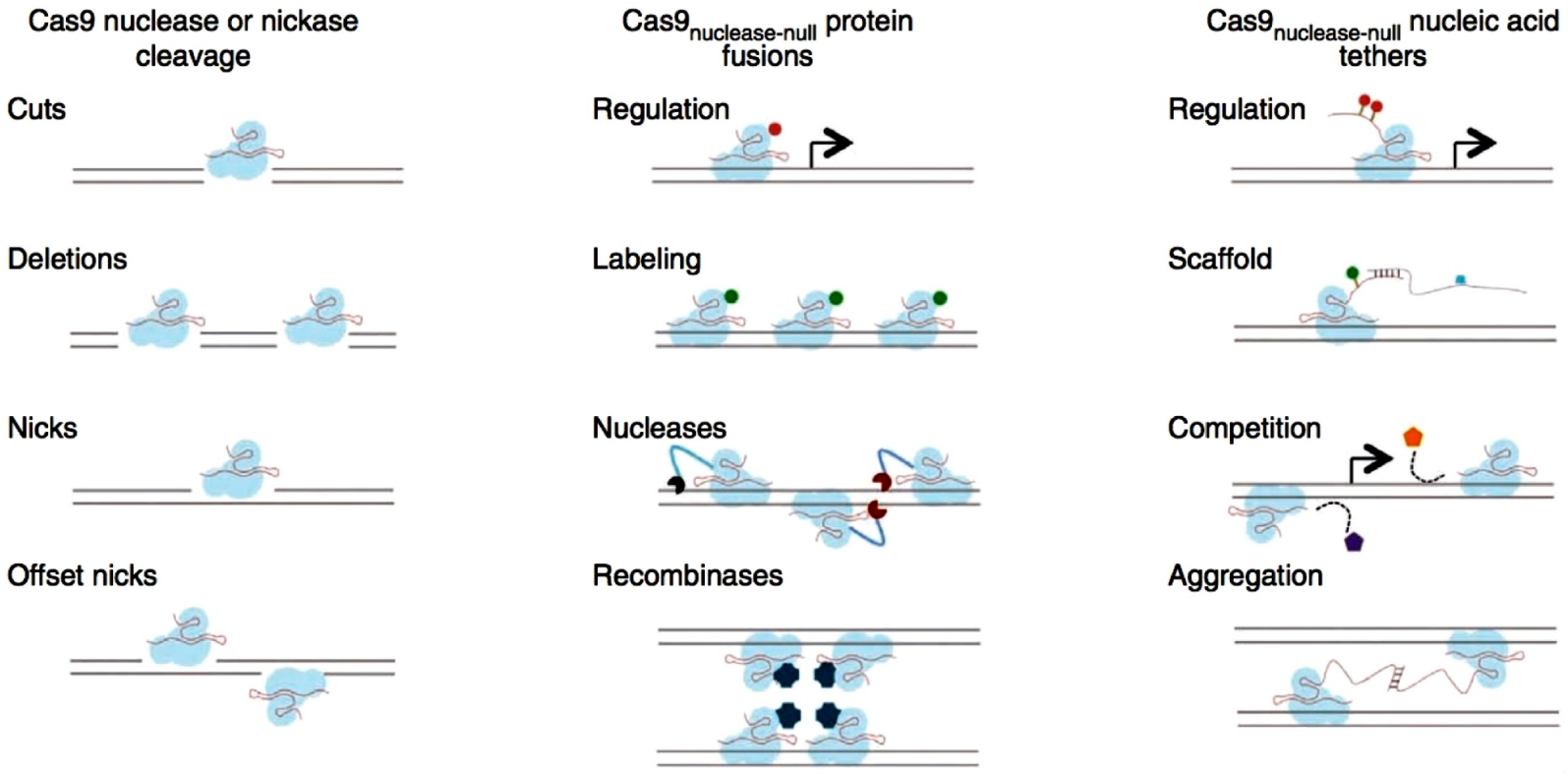


diagnóstico de edición: TIDE

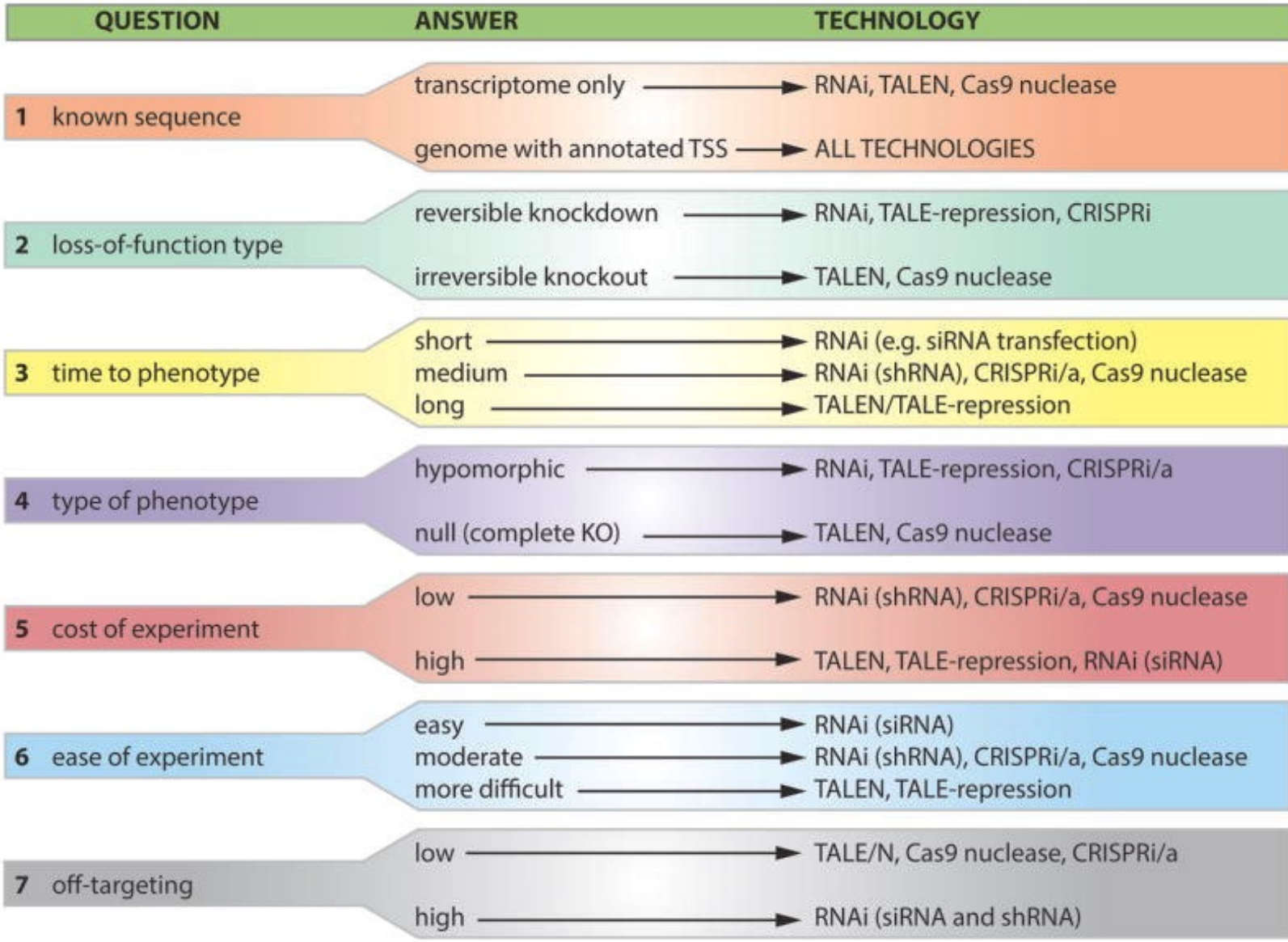
Tracking of Indels by Decomposition



Usos CRISPR/Cas9



¿Qué tecnología de silenciamiento uso?



Bases de datos: métodos de transfección

- https://www.qiagen.com/us/transfectionprotocols/transfectionprotocol/
- https://www.thermofisher.com/ar/es/home/life-science/cell-culture/transfection/transfection-support/transfection-selection-tool.html
- https://knowledge.lonza.com/
- https://www.sigmaaldrich.com/technical-documents/articles/biology/roche/transfection-database.html